ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Electronics and Digital Concepts for Robotics paper 1 · practice a.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct methods (e.g. Karnaugh maps instead of Boolean algebra).

SECTION A

QUESTION 1

1.1 1.1.1 B ✓ 1.1.2 D ✓ 1.1.3 B ✓ 1.1.4 C ✓ 1.1.5 A ✓ 1.1.6 B ✓ 1.1.7 D ✓ 1.1.8 A ✓ 1.1.9 C ✓ 1.1.10 C ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 True ✓ 1.2.3 True ✓ 1.2.4 True ✓ 1.2.5 False ✓ 1.2.6 True ✓ 1.2.7 True ✓ 1.2.8 False ✓ 1.2.9 True ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 A ✓ 1.3.2 F ✓ 1.3.3 G ✓ 1.3.4 E ✓ 1.3.5 C ✓ 1.3.6 B ✓ 1.3.7 H ✓ 1.3.8 J ✓ 1.3.9 I ✓ 1.3.10 D ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 P-type: doped with a trivalent impurity (e.g. boron) ✓, has extra holes – majority carriers are holes (positive) ✓; N-type: doped with a pentavalent impurity (e.g. phosphorus) ✓, has extra free electrons – majority carriers are electrons ✓. (4)

2.2.1 $I_B = I_C \div h_{FE} = 0{,}06 \div 100$ ✓ $= 0{,}6$ mA ✓ (2)

2.2.2 $R_B = (5 - 0{,}7) \div 0{,}0006$ ✓✓ $= 7\,166{,}67\ \Omega \approx 7{,}17\ \text{k}\Omega$ ✓ (3)

2.2.3 A smaller resistor gives more base current ✓, which makes sure the transistor is driven fully into saturation (fully on, low heat) even if h_{FE} is lower than expected ✓. (2)

2.2.4 Microcontroller pin → R_B → base ✓; emitter to 0 V ✓; relay coil between +12 V and collector ✓; diode across the coil, cathode to +12 V (reverse-biased) ✓; common ground between the 5 V and 12 V supplies ✓ (5)

2.3 Any TWO: voltage-controlled – almost no gate current needed from the microcontroller ✓; very low ON resistance, so less heat and higher efficiency ✓; can switch large currents; fast switching for PWM (2)

2.4 It isolates the low-voltage control circuit from the high-voltage circuit using light ✓, so voltage spikes or faults cannot damage the microcontroller / reduces electrical noise ✓. (2)

[20]

QUESTION 3

3.1 (a) $X = A(B + \bar{B})$ ✓ $= A \cdot 1$ ✓ $= A$ ✓ (b) $Y = \bar{A}(\bar{B} + B) + A{\cdot}B$ ✓ $= \bar{A} + A{\cdot}B$ ✓ $= \bar{A} + B$ ✓ (6)

3.2 A NOT gate on input A ✓, then $\bar{A}$ and B into an OR gate giving Y ✓ (2)

3.3.1 Z = 1 for ABC = 011, 101, 110, 111; Z = 0 for 000, 001, 010, 100 ✓✓✓✓ (1 mark per two correct rows) (4)

3.3.2 $Z = A{\cdot}B + A{\cdot}C + B{\cdot}C$ ✓✓✓ (3)

3.3.3 Three two-input AND gates (AB, AC, BC) ✓✓ feeding a three-input OR gate ✓ (3)

3.4 NOT: both inputs of a NAND gate joined together ✓; AND: a NAND gate followed by a NAND gate wired as a NOT ✓ (2)

[20]

QUESTION 4

4.1 Truth table ✓✓✓✓: S = 0, R = 0 → no change (memory); S = 1, R = 0 → Q = 1 (set); S = 0, R = 1 → Q = 0 (reset); S = 1, R = 1 → not allowed / invalid ✓ (5)

4.2 The button is connected to D ✓; when a clock pulse (or a second signal) occurs, Q takes the value of D ✓ and keeps it (stores it) until the next clock pulse, even after the button is released ✓. (3)

4.3 Asynchronous: only the first flip-flop gets the clock; each flip-flop clocks the next ✓, so outputs change one after the other with small delays (ripple) ✓. Synchronous: all flip-flops share the same clock ✓ and change at the same time – faster and no ripple errors ✓. (4)

4.4 Four ✓; three flip-flops can only count to 7 ($2^3 = 8$ states) ✓, four give 16 states, and the counter is reset after 9 ✓. (3)

[15]

QUESTION 5

5.1.1 $\text{Resolution} = 5 \div 1\,024$ ✓ $= 4{,}88\ \text{mV}$ ✓ (2)

5.1.2 $\text{Value} = 1{,}25 \div 5 \times 1\,024$ ✓ $= 256$ ✓ (accept 255 with 1 023 used) (2)

5.1.3 $V = 818 \times 4{,}88\ \text{mV}$ ✓ $= 3{,}99\ \text{V}\ (\approx 4\ \text{V})$ ✓ (2)

5.2 $V_{out} = 5 \times 10 \div (10 + 1)$ ✓✓ $= 4{,}55\ \text{V}$ ✓ (3)

5.3 Hardware: an RC filter (resistor and capacitor) or a Schmitt trigger smooths the signal ✓✓; software: after detecting a press, wait a short time (e.g. 20–50 ms) and read the input again before accepting it ✓ (3)

5.4 Analogue sensor gives a continuously varying voltage ✓, e.g. LDR, potentiometer, temperature sensor ✓; digital sensor gives only ON/OFF (1/0) or a digital code ✓, e.g. limit switch, IR obstacle module, encoder (3)

[15]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE